Which Short Run? Whose Money? Closure Choice in Input–Output Models of the Socio-Ecological Transformation*

Jakob Kapeller[1] and Franz Scharnreitner[2]

[1]Corresponding author, Institute for Socio-Economics, University of Duisburg-Essen and Institute for Comprehensive Analysis of the Economy (ICAE), Johannes Kepler University Linz, jakob.kapeller@uni-due.de

[2]Institute for Comprehensive Analysis of the Economy (ICAE), Johannes Kepler University Linz

October, 2026

Abstract

[Abstract to be drafted.]

Input-output models are a widely used tool for estimating the economic effects associated with societies moving towards carbon-neutrality or, more fundamentally, aiming to achieve a socio-ecological transformation. While this practice documents that such models are perceived as conceptually useful and empirically relevant, such approaches often take input-output models as a supposedly neutral analytical device. Against this background, this paper revisits how paradigmatic perspectives matter for input-output modeling. It shows that such perspectives and the associated analytical differences have a substantial impact on estimated effects at various levels. This observation indicates how modern applications of established models are actually haunted by ghosts from the past that reflect long-standing debates, and this has impact on the interpretation of a great variety of published results. Using the case of an archetypal green transformation policy, the decarbonization of the German housing sector, we show how different axiomatic setups imprint on estimates of the economic effects of ecologically motivated transformation efforts. By doing so, we hope to provide applied researchers with better guidance for reflecting on how the diversity of available interpretations of input-output models impacts

*This paper has benefited from funding provided by the *Hans Böckler Foundation* under grant number 2021-544-2, by the *Linz Institute for Transformational Change (LIFT_C)* under grant number 2022-1-114 as well as by *Dezernat Zukunft*. Moreover, we thank Lukas Cserjan and Bernhard Schütz for their extensive feedback as well as David Baqaee, Anna Hornykewycz, Stefan Steinerberger, Rafael Wildauer and Jan David Weber for their responses to our queries and the comments on specific aspects of this manuscript. All remaining errors are our own.

on their problem at hand.

Keywords: input–output modeling, general equilibrium, closure choice, green investment, socio-ecological transformation

JEL-Codes: C67, C68, D57, E24

1 Introduction

Input-output models are an established tool for estimating the economic effects of structural policy choices related to public investment, industrial policy or labor market trends. Against this background it does not come as a surprise that input-output models are also widely used to estimate the economic effects of additional investment undertaken in the course of a socio-ecological transformation aiming to render social provisioning processes carbon-neutral (Creutzig et al., 2018; Hornykewycz et al., 2025; He et al., 2023). However, it is well known that the results obtained from such models depend crucially on the underlying axiomatic setup. Variations in outcomes emerging from variations of the chosen setup are seldom reported in the literature on transformation efforts that resorts to input-output modeling. Moreover, and more importantly, little attention is paid to the fact that the axiomatic setup of input-output models reflects economically important contextual arguments. In this paper we recapitulate that substantial variation in outcomes is due to a specific part of the axiomatic setup, namely the different closure choices. For research on socio-ecological transformation efforts it is important to recognize that such closure choices are not mere technical decisions, but carry substantial economic content; they capture – or presuppose – certain contextual arguments, assumptions about the current state of the economy, the treatment of the future and the exact form in which some policy proposal is executed in practice.

Closure choices carry substantial economic content as they make (implicit) assumptions about the context and conditions under which the suggested policy is implemented. This observation resonates with arguments on the importance of socio-historical and institutional context in economics (Lawson, 1985; Dobusch and Kapeller, 2012) as neatly summarized by Keynes's quip that "economics is a science of thinking in terms of models joined to the art of choosing models which are relevant to the contemporary world" (Keynes, 1938). In this spirit our argument is also grounded in the observation that closure choices often reflect traditional divides in economic theory and policy so that, in effect, old ghosts haunt modern applications of input-output models, which have become a main workhorse in modeling socio-ecological transformation efforts (Hornykewycz et al., 2025; Creutzig et al., 2018).

It is well known that such paradigmatic cleavages matter greatly for input-output analysis: while the question whether or to what relative extent output or prices react to a change in final demand (scarcity vs. output-adjustment) is most prominent as it sits at the core of the division between computable general equilibrium models (CGEs) and the more traditional linear extrapolations of production structures based on the Leontief Inverse (SOURCES, CLASSICS PAPER BY

LEONTIEF??). Moreover, a series of other conceptual questions come into view when reflecting on paradigmatic cleavages in input-output modeling: choices on production technology (Cobb Douglas vs. Leontief; Labini, 1995; Klump, McAdam, and Willman, 2012), financing (external vs. internal; Baqaee and Farhi, 2019; Baqaee and Farhi, 2022), policy implementation (incentivising private actors vs. state-led action) or technological change (exogenous vs. endogenous; Kaldor, 1961) are all examples of such foundational questions, where disagreement translates into a structural wedge on received estimates and projections. In this paper we show that such conceptual questions have a substantial impact on the estimated effects of socio-ecological transformation efforts and which of these questions have the strongest impact on final results. To substantiate and illustrate this claim we use the case of a well-understood green transformation policy, the decarbonization of the German housing sector, and show how different axiomatic setups imprint on final estimates. By doing so, we hope to provide applied researchers with better guidance for reflecting on how the diversity of available interpretations of input-output models impacts on their problem at hand.

In what follows we will specifically focus on two types of closures, namely labor market closures and financing closures, which can be mapped onto a configuration space with two categorical axes. In this space one axis goes from fundamental scarcity of factors (i.e. fully utilized factor markets) to flexible output adjustment (in the spirit of the Keynesian multiplier), which leads to fundamentally different conceptions on the short run. The second axis focuses on financing and goes from the assumption of immediate refinancing (additional expenditures lead to cuts elsewhere) to assuming some form of external financing that relieves the contemporary economy from immediate refinancing needs. These axes are conceptually captured by Table 1 below, which gives a first, simplified representation of the cleavages under study.

	Labour-market closure	
	Scarce factor supply Computable General Equilibrium (CGE)	**Real wage fixed** Leontief-Inverse and Output adjustment
Repurposing consumption (F1)	pure reallocation	the change follows the input–output structure
Refinancing via taxes (F2)	crowding out	balanced-budget multiplier (Haavelmo)
External finance (F3)	welfare loss and external debt	output expansion and external debt

Table 1: A simplified map of two main closure axes in input-output modeling.

The table is deliberately kept simple: it collapses the labour-market axis into two poles and is silent on magnitudes. The paper's central object is the finer grid behind it – a five-by-three closure

matrix (five labour-market closures crossed with the three financing regimes above), evaluated on a single, empirically informed green-investment impulse. Running that grid illustrates our main message, namely there is no closure-free multiplier. Every estimate of a transformation programme presupposes a short run and a financing regime, and that choice is not a technicality – it is first-order for the magnitude, and occasionally for the sign, of the result. The matrix then identifies where closure choice is redundant (the fixed-wage rows GAMMA and DELTA, which coincide exactly, mimicking the basic intuition of Table 1; BETA is the sectoral-wage form and is a distinct economy) and where it decides the answer.

The remainder of this paper is structured as follows. Section 2 connects transformation research to input–output modeling and introduces the green-investment impulse. Section 3 develops the two primitive questions – the nature of the short run and the role of money – and the simplified matrix. Section 4 presents the common kernel, the five labor-market closures and the three financing regimes. Section 5 runs the programme through the matrix, and Section 6 considers shocks beyond demand-only impulses. Section 7 discusses the results and Section 8 concludes. Proofs and calibration details are collected in the appendices.

How to read this paper. Beginners: Sections 1, 2 and 5. Experts: Sections 3, 4 and 6. Proofs are in Appendix A.

2 From transformation research to input–output models

Modest proposals for decarbonization initiatives and large-scale suggestions for fundamental transformation both imply a repurposing of spending, either by private agents, the public sector or an interaction of both. Hence, such programmes share a similarity with traditional public spending programs (e.g. Pollin, 2015; Mazzucato, 2013), which also invite reflection on the economic impacts of such repurposed or additional spending. Classic concepts that come to mind in this context are the Keynesian cross (Keynes, 1936) (implicitly building on the idea of external financing), the Haavelmo theorem (Haavelmo, 1945) (which just states that the state – in contrast to households – spends all of its income) or the idea of a "crowding out" of public spending (Blinder and Solow, 1973) (similar to Haavelmo, but typically with adverse effects due to less investment).

Several contributions in the recent literature (Mazzucato, 2013; Mazzucato, 2018; Pollin, 2015) exploit this similarity in various ways arguing that public spending is a functional means or necessary

requirement for decarbonization and socio-ecological transformation. Aside from such more technocratic takes, a series of authors identify political synergies on this basis, where public spending alleviates both, environmental degradation as well as social exclusion, and, hence, serves as a foundational keystone for proposals on a just transition (Newell and Mulvaney, 2013). Moreover, analytical approaches to input-output models sometimes emphasize their capability to identify structural dependencies and critical junctures, allowing for assessing the resilience and versatility of economies in times of intense transitions (Farmer, 2024; Cimini, Mastrandrea, and Squartini, 2021; Holý and Šafr, 2022).

Finally, as a traditional application of input-output models in the spirit of a simple application of a Leontief inverse assumes an output expansion analogous to the Keynesian cross, this basic intuition of synergy can also be found on a more analytical level – albeit only when taking, maybe unconsciously, a specific paradigmatic perspective. Given their analytical and conceptual advantages, their wide applicability and their resonance with contemporary political challenges it does not come as a surprise that also many applied studies in transformational research make use of input-output models to assess the potential impacts of such policies (Creutzig et al., 2018; Hornykewycz et al., 2025; He et al., 2023).

Our exemplary application goes to the heart of these debates by focusing on the key issue in transformation debates, namely the decarbonization of the residential sector. By taking Germany as an example we focus on a major European economy that is marked by a strong path dependence towards the use of fossil energy sources, especially in the residential sector, but also in energy production, which is key for providing alternative means of provisioning basic housing services. We start from a concrete estimate for additional investment requirements in Germany to arrive at a carbon-neutral residential sector by 2050 (Hornykewycz et al., 2025). Such an effort would require more than doubling spending on renovation and increase annual expenditures on this by about 58 bn per year on average (in 2023 prices), focused on a small subset of all sectors.[1] In 2019 prices, the base year of the input–output table, this corresponds to 40.3 bn, or 1.33% of GDP. By taking this number – and the estimated sectoral distribution of spending needs – as our vantage point, we anchor our illustration in an empirically and politically relevant case that is a valid example for the kind of "repurposing of existing expenditures" that a socio-ecological transformation would entail.

[1] The following sectors are subject to demand shock (with expected relative change given in brackets): Glass and glassware (+50.3%), ceramic products and processed stones (+37.3 %), specialized construction work (+27.8%), rubber and plastic products (+13.4%), chemicals and chemical products (+3.7%), machinery (+1.2%) and electrical equipment (+1.0%).

3 Two primitive questions, a simplified matrix and some complications

As already indicated, key points of disjuncture in input-output modeling can be mapped onto classic cleavages separating different traditions of economic thought. Such traditions can often be characterized in terms of the basic analytical ideas – or "rivers" in the history of economic thought (SCREPANIT/ZAMGANI with p.) – they endorse. Where these basic ideas have a different emphasis or contradict outright, a paradigmatic cleavage is in sight (Dobusch and Kapeller, 2012). Two such classic fields of tension in economic thinking also lie at the heart of different approaches to input-output modeling: the nature of the short run and the role of money and credit, as already captured by Table 1.

When mapping the rough distinctions from our introductory table more specifically to the realm of input-output models, we find that different labor market closures are the key entry point for different understandings of the short run: if labor is tight or fully utilized, increased monetary demand will increase wages and prices as capacity constraints bind as computable general equilibrium models (CGEs) typically suggest. Conversely, if free factors of production are available, additional expenditures will lead to an expansion of employment and, correspondingly, output. This classic output adjustment closure (**Taylor1990**) is what one gets from a simple linear extrapolation of the Leontief inverse. Both visions can be unified in one analytical apparatus (see **Robinson.2006** and Section 4), just differentiated by different assumptions on what is exogenous or endogenous when monetary expenditures increase. The two poles are the *standard CGE closure*, in which one flexible wage clears the labour market at full employment and only relative prices move, and the *Leontief closure*, in which fixed coefficients and a fixed real wage leave prices pinned so that output absorbs demand. While the former is the obvious neoclassical standard model, the latter reflects the intuition of a multiplier and a notion of economic planning that shows greater resonance with (Post-)Keynesian, radical and evolutionary approaches (SOURCES????).

Financial closures on the other hand encode prior beliefs about the role of our monetary system and the nature of money. As input-output models are typically conceived as static, operating outside of time (see however AUSNAHMEN ZITIEREN??), there is no borrowing against the future – additional expenditure cannot be externally financed in the conventional sense, but has to be repaid within the same period (see however AUSNAHMEN ZITIEREN??). Hence, while a traditional, simplified application of a Leontief inverse would suggest some external stimulus, exactly such an

external force is at odds with the internal consistency of the model. In other words, canonical input-output models implicitly build on an exogenous view on money and restrict the application of ideas on endogenous money creation as it lacks a any framework to conceptualizethe intertemporal role of debt.

Refinancing can then occur in three ways: by households – the repurposing of consumption (F1) –, via the state – refinancing through increased taxation (F2) –, or by someone willing to take on foreign debt to provide an external stimulus (F3). Such debt provides an accounting analogue that can be associated with diverse form of external finance within an input–output framework without explicitly violating its internal consistency properties. Mapping these two dimensions onto specific modeling approaches allows for recasting Table 1 in terms of specific constellations of closure summarized by Table 2 below.

	Short run: standard CGE one flexible wage, labour mobile, full employment	**Short run: standard Leontief** fixed real wage, fixed coefficients, output absorbs
Financing: F1 repurposing of consumption	$L_i = L_i^{\mathrm{cm}}(p, y_i, w), \quad \sum_i L_i = \bar{L}$ $\sum_i p_i c_i^h = E^h$	$w/\Pi(p) = 1, \quad (\theta, \epsilon, \sigma) \to 0^+$ $\sum_i p_i c_i^h = E^h$
Financing: F2 refinancing via taxes	$L_i = L_i^{\mathrm{cm}}(p, y_i, w), \quad \sum_i L_i = \bar{L}$ $T(p) = \sum_i p_i g_i$	$w/\Pi(p) = 1, \quad (\theta, \epsilon, \sigma) \to 0^+$ $T(p) = \sum_i p_i g_i$
Financing: F3 foreign debt	$L_i = L_i^{\mathrm{cm}}(p, y_i, w), \quad \sum_i L_i = \bar{L}$ $B_{\mathrm{gov}} = \sum_i p_i g_i$	$w/\Pi(p) = 1, \quad (\theta, \epsilon, \sigma) \to 0^+$ $B_{\mathrm{gov}} = \sum_i p_i g_i$

Table 2: The simplified matrix: the two standard short-run closures crossed with the three financing closures. Each cell states the two closure conditions that define that economy — one for the short run (labour market), one for the financing of the programme. Notation follows Section 4; $\Pi(p)$ is the consumption price index, E^h household consumption expenditure, $T(p)$ the programme tax and B_{gov} the externally booked programme cost. "Standard CGE" is the paper's ALPHA row — one wage clears, labour is mobile, employment is full; "standard Leontief" is the paper's DELTA row — a fixed real wage combined with the Leontief limit of the CES core, so that quantities absorb demand. The finer grid of Section 4 fills in the labour-market closures between these two poles.

While the crosswise recombination of these different options on closure choices already gives rise to some complexity, the underlying choice menu is nonetheless somewhat rough and – in the case of the short-run – simplistically dichotomous. Hence, it comes as no surprise that further established variations of closure setups exist that try to address this point. An obvious starting point from a mainstream perspective to introduce greater variety in labor market outcomes is by explicitly referring to a leisure-income trade-off, which leads to the canonical *elastic labor supply model*: Assuming that the substitution effect dominates, higher wages will attract additional labor so that capacity constraints do not bite with full force. In the *elastic labour supply* closure the fixed labour endow-

ment is replaced by a sector-specific labor supply curve, $L_i^s = \bar{L}_i \, [\, (w_i/\Pi(p))/(w_i^0/\Pi^0) \,]^{\eta_s}$, with one common elasticity η_s. Two changes of functional form follow: labour supply becomes *behavioural* rather than a quantity restriction or a price pin, and the single aggregate labour market becomes N sectoral markets with N wages.[2]

While this modification brings greater responsiveness on the side of employment other variations in the mainstream literature assume forms of rigidity that go even beyond the conventional CGE labor market closure. A prominent recent example is given by Baqaee and Farhi (2019), who introduce a "no-reallocation" case as a baseline scenario to articulate the view that sectoral productivity shocks might impact on aggregate outcomes in a non-linear way. Relative to Table 2 this is not a column but a more rigid, sectoral extension of the labour axis: it projects the standard CGE column's aggregate restriction on a sectoral level, i.e., $\sum_i L_i = \bar{L}$ (fixed employment levels for all sectors, so "no-reallocation") and correspindingly relies on sectorally flexible wage w_i. Formally, the transition from the standard CGE model to the no-reallocation model is governed on the production side, where η represent the mobility of labor across sectors, whith $\eta = 0$ for the no-reallocation case.

In the context of our case of application – a socio-ecological transformation that is tied to sector-specific shifts or injections in final demand – such a baseline setup is especially pessimistic as it assumes labor market rigidity across sectors. As an effect, sectors with strong demand shocks are then hopelessly overwhelmed as workers from other sectors cannot "reallocate" to where capacity is needed. This result, however, resonates with intuition that additional rigidities on labor markets will complicate transformation efforts building on exactly such a reallocation of labor.[3]

Technically the *Leontief closure* named here can be represented with the very same analytical apparatus as the conventional CGE model: with substitution elasticities taken to their Leontief limits, fully flexible production factors and the real wage pegged (7), prices then follow from constant costs and final demand determines output. This is the corner most applied transformation studies work with, usually without emphasis. Given this restrictive setup the *Leontief closure* can itself be conceived as corner solution of a broader set of related approaches that retain the basic idea that "quantities adjust" without also committing to a Leontief production function. In their basic form these *fixed (real) wage* models assume constant real wages (AG: EQUATION HERE), which pins down priced and leaves the quantity margin free. They stand in the same tradition as the Leontief

[2]The latter assumption is not strictly necessary, but relevant for our case of application which typically implies shocks with rather heterogenous impact on sectors.

[3]In this case assumptions deviate from the neoclassical baseline, but strengthen the general laissez-faire policy stance associated with its standard closure. Also note that related work on rigidities in capital or financial markets also exists, but is beyond the scope of this paper (see, however, (SOME SOURCES))

inverse: it is the (more general) closure through which the Keynesian multiplier intuition enters the kernel (see also section 4). ibed: there the allocation is frozen sector by sector and the

Observing these additional models and corresponding closure variations moves our view on the short run from the manicheistic scenario formulated in table 2 to a more nuanced exposition of different conceptions of the short run: the *no-reallocation approach* imposes additional rigidity on labor markets, *elastic labor supply models* allow for introducing behavioral responses and the *fixed real wage models* try to generalize the scope of the output adjistment view. Moreover, the first two of these approaches are articulated on a sectoral level and the latter can be extended to one. In what follows we suggest to take the now resulting portfolio of closure choices as means to illustrate the consequences of different closure assumption – and, hence, paradigmatic perspectives – on results when study the impacts of a socio-ecological transformation ny means of input-output models.

Before we map these five closures in a monumental table further below (REF!) it makes sense to point out the sometimes surprising overlaps that arise for certain characteristic parameter values. For one, the no-reallocation model differs from the conventional CGE model only by exactly this "no-reallocation" assumption, wherese choosing a reallocation term $\eta = 1$ instead of $\eta = 0$ wold lead to full convergence with the conventional case (as the sectoral level ceases to matter with "full reallocation"). For another, the elastic labor supply model – being a curve of solutions rather than a simple corner – is able to mimic the outcome of the no-reallocation model by assuming that labor market participants lack any flexibility (by setting $eta_s = 0$), as well as that of the *fixed real age* model by taking the inverse position (letting $eta_s \to \text{inf}$). Hence, the elastic labor supply model can be said to sit at the centre of the interparadigmatic bridge between neoclassical interpretations based on factor scarcity and more Keynesian multiplier-style interpretations based on extrapolation via the Leontief inverse (**Robinson.2006**; **Taylor1990**).[4] Finally, as already indicated, the *fixed real wage* closure is equivalent to the classic Leontief result, when production structures are assumed to be rigid.

==========

4 One kernel, five closures, three financings

We examine how different closures have different effects. For this we choose five different labor closures and three financing closures that reflect different assumptions about the economy. The

[4] Also, the elastic labor supply model collapses to te conventional CGE in the absence of sectoral heterogeneity.

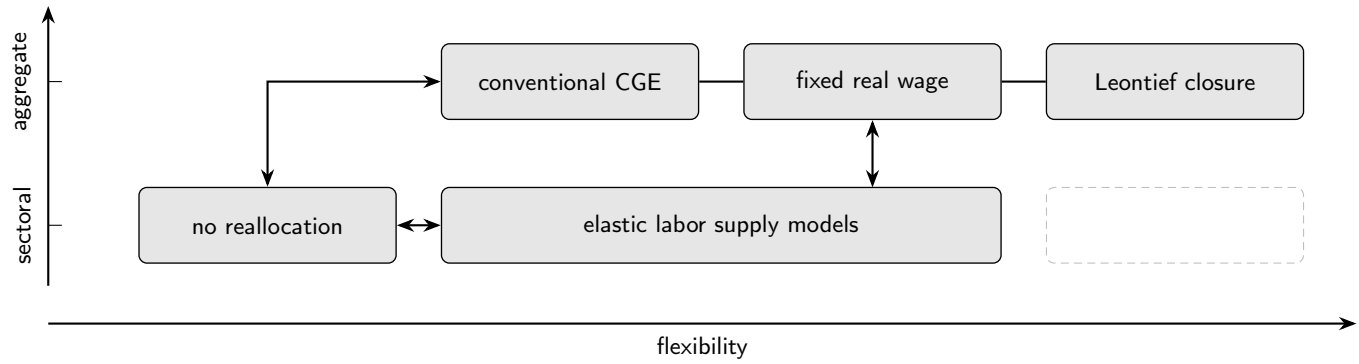


Figure 1: A map of closure choices along the short-run (labour-market) axis. Columns order the closures by increasing flexibility; the aggregate row collects the single-wage closures, the sectoral row the sector-level ones. The double-width box holds the elastic-labour-supply family, tied by short double arrows to the no-reallocation and fixed-real-wage corners it interpolates between; the right-angled double arrow marks the reallocation margin from no reallocation up to the conventional CGE closure.

labour closures specify how wages and employment adjust. **BF** fixes employment in each sector at its baseline level and lets sector-specific wages adjust. **ALPHA** allows labour to move between sectors at one flexible wage while keeping total employment fixed. **BETA** lets sectoral employment respond to each sector's real wage, deflated by the consumption price index, with a common supply elasticity but separate wages. **GAMMA** fixes the common real wage and lets employment adjust without a cap. **DELTA** applies GAMMA's wage rule in the Leontief limit, with fixed proportions in production and consumption; it is a corner rather than a separate labour-market mechanism.

The financing closures specify how programme-related demand is accommodated and who bears its cost. **F1** redirects household spending towards programme sectors by changing preferences: larger spending shares there are paid for by smaller shares elsewhere within the household budget, without additional public purchases. **F2** adds public investment financed by a household tax equal to the programme's current-price cost. **F3** adds the same public-investment bundle with external financing and no additional domestic programme tax. This external financing is an accounting counterpart, not a monetary mechanism.

4.1 The five labour closures

Let L_i denote employment in sector i, $\bar{L}_i$ its baseline value, and $\bar{L} = \sum_{i=1}^{N} \bar{L}_i$ total baseline employment. Sectoral wages are w_i; with mobile labour they equal one common wage w. The real wage is

deflated by the consumption price index

$$\Pi(p) = \left(\sum_{i=1}^{N} \omega_i p_i^{1-\sigma} \right)^{1/(1-\sigma)}, \qquad \sum_{i=1}^{N} \omega_i = 1, \tag{1}$$

with calibrated weights ω_i and the Cobb–Douglas limit at $\sigma = 1$. Subscript 0 denotes the baseline. All closures use the same cost-minimizing labour demand,

$$L_i^{\mathrm{cm}}(p, y_i, w_i) = \alpha_i A_i^{\epsilon-1} y_i \left(\frac{p_i}{w_i} \right)^{\epsilon}. \tag{2}$$

Here y_i is gross output, A_i productivity and α_i the calibrated labour share. The elasticity ϵ governs substitution between labour and the intermediate composite. The closures differ in the wage and labour-supply restrictions imposed on this demand.

BF: immobile labour. The BF row fixes employment in every sector and solves a sector-specific wage to clear each labour market:

$$L_i = L_i^{\mathrm{cm}}(p, y_i, w_i) = \bar{L}_i, \qquad i = 1, \ldots, N, \qquad \eta = 0. \tag{3}$$

Thus $\sum_i L_i = \bar{L}$, but wages may differ across sectors. The allocation parameter η selects only the endpoints $\{0, 1\}$; its mobile endpoint $\eta = 1$ is ALPHA below, not an additional closure.

ALPHA: mobile labour at full employment. Labour reallocates across sectors at one flexible wage, while total employment remains fixed:

$$L_i = L_i^{\mathrm{cm}}(p, y_i, w), \qquad \sum_{i=1}^{N} L_i = \bar{L}, \qquad \eta = 1. \tag{4}$$

BETA: elastic sectoral labour supply. We use the uniform-elasticity sectoral form: each sector has its own wage and supply curve, with one common labour-supply elasticity η_s:

$$L_i = L_i^{\mathrm{cm}}(p, y_i, w_i) = \bar{L}_i \left[\frac{w_i/\Pi(p)}{w_{i0}/\Pi_0} \right]^{\eta_s}, \qquad \eta_s \geq 0, \qquad i = 1, \ldots, N. \tag{5}$$

Equal supply elasticities do not impose equal wages: N separate labour markets determine w_i. Here η_s is a labour-supply elasticity, not the allocation parameter η; setting $\eta_s = 0$ gives exactly the immobile BF row. The calibrated real-wage anchors are $w_{i0}/\Pi_0 = 1$. The deflator remains

the common consumption price index, not the sector's own goods price p_i. The supply curves are reduced forms without a labour–leisure income effect. The headline panels use $\eta_s = 0.5$; the elasticity ladder and heterogeneous sectoral variants are reported in Appendix C. The single-wage BETA is retained there only as a distinct benchmark; its zero-elasticity limit is ALPHA, not BF.

GAMMA: fixed real wage. The real wage is pegged at its baseline value, while employment is endogenous and uncapped:

$$\frac{w}{\Pi(p)} = \bar{w} = 1, \qquad L_i = L_i^{\mathrm{cm}}(p, y_i, w), \qquad \eta = 1. \tag{6}$$

Unlike ALPHA, this closure does not impose $\sum_i L_i = \bar{L}$. It is a two-sided wage peg, not a one-sided wage floor with unemployment rationing. On the demand-only baseline branch, $\Pi(p) = 1$ and the implemented wage pin is $w = 1$.

DELTA: the Leontief corner. DELTA retains GAMMA's wage and employment rule while taking the substitution elasticities to their Leontief limits:

$$\frac{w}{\Pi(p)} = \bar{w} = 1, \qquad \eta = 1, \qquad (\theta, \epsilon, \sigma) \longrightarrow (0^+, 0^+, 0^+). \tag{7}$$

Here θ governs substitution between intermediate inputs and σ substitution in consumption. DELTA is therefore a corner of GAMMA with fixed-coefficient technology, rather than an independent labour-market mechanism.

4.2 The three financing closures

Each labour closure is crossed with three ways of accommodating the same programme scale and sectoral incidence. Let $g_i = kG_0\psi_i$ denote the reference programme bundle, where G_0 is its baseline-price value, $k \geq 0$ scales the programme, and $\psi_i \geq 0$ with $\sum_i \psi_i = 1$ describes its sectoral incidence. F2 and F3 add this bundle to final demand; F1 instead uses its scale and incidence to redirect household spending.

Baseline government purchases g_i^G remain tax-financed in all three cases. Define the wage bill and their current-price tax cost as

$$W = \sum_{i=1}^{N} w_i L_i, \qquad T_G(p) = \sum_{i=1}^{N} p_i g_i^G. \tag{8}$$

With a common wage, $W = w\sum_i L_i$. Let E denote after-tax household resources, including the net external transfer F, and let s be the calibrated saving rate. Household consumption expenditure is $E^h = (1-s)E$. Gross household purchases c_i^h exhaust this budget:

$$\sum_{i=1}^{N} p_i c_i^h = E^h, \qquad c_i^{\text{dom}} = (1-m_i)c_i^h, \tag{9}$$

where m_i is the sectoral import margin. Only the domestic content enters domestic goods-market clearing; likewise, the programme contributes $(1-m_i)g_i$ under F2 and F3.

F1: budget-neutral preference reallocation. F1 changes the household's CES weights without adding a public-demand bundle. For positive preference shifters d_i, the weights and gross household demand become

$$\widetilde{\omega}_i = \frac{\omega_i d_i}{\sum_j \omega_j d_j}, \qquad \sum_i \widetilde{\omega}_i = 1, \tag{10}$$

$$c_i^h = E^h \frac{\widetilde{\omega}_i p_i^{-\sigma}}{\sum_j \widetilde{\omega}_j p_j^{1-\sigma}}, \qquad E = W - T_G(p) + F. \tag{11}$$

The programme-scale tilt is $d_i = 1 + kG_0\psi_i^+/c_{i0}^h$ in sectors with positive baseline household purchases, and $d_i = 1$ otherwise; ψ^+ is the programme incidence restricted to those sectors and renormalized to sum to one. The CES normalizer preserves (9) at every price vector. Thus increased spending shares in programme sectors are paid for by reduced shares elsewhere, conditional on the household budget; equilibrium income need not remain unchanged. F1 is a demand-composition experiment, not an autonomous investment injection or multiplier.

F2: tax-financed public investment. F2 adds the real programme bundle g and levies an additional household tax equal to its value at equilibrium prices:

$$T(p) = \sum_{i=1}^{N} p_i g_i, \qquad E = W - T_G(p) - T(p) + F, \qquad B_{\text{gov}} = 0. \tag{12}$$

Total domestic tax revenue $T_G(p) + T(p)$ therefore balances the purchases $g^G + g$, priced at current rather than baseline prices. The tax is implemented as the endogenous wage-income rate $\tau = [T_G(p) + T(p)]/W$; with one representative household its burden is effectively lump-sum. The programme has an explicit domestic budget counterparty and reduces household resources directly by $T(p)$.

F3: externally financed public investment. F3 adds the same real bundle but imposes no additional domestic programme tax. Its current-price cost is booked as external programme financing:

$$B_{\text{gov}} = \sum_{i=1}^{N} p_i g_i, \qquad E = W - T_G(p) + F. \tag{13}$$

The booking B_{gov} is distinct from the net external transfer F entering household resources after tax. The latter is solved endogenously in BF, ALPHA and both BETA forms, but is fixed at zero in GAMMA and DELTA. With no unfinanced demand injection, all closures clear all N goods markets and satisfy the external-account identity

$$S + T_{\text{int}} + M - (I + X) = F + B_{\text{gov}}, \qquad S = sE, \tag{14}$$

where I is non-programme investment, X exports, M total imports (including intermediate and programme imports), and T_{int} the separately booked product-tax leak on intermediate use; all are current-price values. Here $B_{\text{gov}} = 0$ under F1 and F2, so the net external position is $F + B_{\text{gov}}$, not the programme cost alone. F3 is an accounting open-economy closure, not a monetary mechanism: the static model cannot distinguish external borrowing from a transfer and does not model interest payments, repayment or exchange-rate adjustment.

4.3 Why only three economies

4.4 Presentation rule

5 Application: the programme through the matrix

5.1 Aggregate results

5.2 Sectoral results

5.3 The financing wedge

5.4 Robust vs closure-sensitive

6 Beyond demand-only shocks

6.1 The equivalences and their scope

6.2 The sectoral family

6.3 The supply-shock arm

6.4 The no-go corollary and the BF pole

7 Discussion

8 Conclusion

Data availability

The input–output data for Germany employed in this paper is publicly available (see the Destatis input–output accounts cited in Appendix D).

Funding

This paper has benefited from funding provided by the *Hans Böckler Foundation* under grant number 2021-544-2, by the *Linz Institute for Transformational Change (LIFT_C)* under grant number 2022-1-114 as well as by *Dezernat Zukunft*.

Conflict of interest

The authors report no conflicts of interest associated with the content of this paper.

References

Baqaee, D. R. and E. Farhi (2019). "The Macroeconomic Impact of Microeconomic Shocks: Beyond Hulten's Theorem". In: *Econometrica* 87.4, pp. 1155–1203.

– (2022). "Supply and Demand in Disaggregated Keynesian Economies with an Application to the COVID-19 Crisis". In: *American Economic Review* 112.5, pp. 1397–1436.

Blinder, A. S. and R. M. Solow (1973). "Does Fiscal Policy Matter?" In: *Journal of Public Economics* 2.4, pp. 319–337.

Cimini, G., R. Mastrandrea, and T. Squartini (Aug. 31, 2021). *Reconstructing Networks*.

Creutzig, F. et al. (2018). "Towards demand-side solutions for mitigating climate change". In: *Nature Climate Change* 8.4, pp. 260–263.

Dobusch, L. and J. Kapeller (2012). "Heterodox United vs. Mainstream City? Sketching a Framework for Interested Pluralism in Economics". In: *Journal of Economic Issues* 46.4, pp. 1035–1058.

Farmer, J. D. (2024). *Making Sense of Chaos: A Better Economics for a Better World*. New Haven, CT: Yale University Press.

Haavelmo, T. (1945). "Multiplier Effects of a Balanced Budget". In: *Econometrica* 13.4, pp. 311–318.

He, X. et al. (2023). "The role of renewable energy investment in tackling climate change concerns: Environmental policies for achieving SDG-13". In: *Sustainable Development* 31.3, pp. 1888–1901.

Holý, V. and K. Šafr (July 27, 2022). *Disaggregating Input–Output Tables by the Multidimensional RAS Method: A Case Study of the Czech Republic*. URL: `http://arxiv.org/abs/1704.07814` (visited on 03/12/2024). Pre-published.

Hornykewycz, A. et al. (2025). "Carbon neutrality in the residential sector: a general toolbox and the case of Germany". In: *npj Climate Action* 4.1, p. 31.

Kaldor, N. (1961). "Capital Accumulation and Economic Growth". In: *The Theory of Capital: Proceedings of a Conference held by the International Economic Association*. Ed. by D. C. Hague. London: Palgrave Macmillan UK, pp. 177–222.

Keynes, J. M. (1936). *The General Theory of Employment, Interest and Money*. London: Macmillan.

– (1938). "Letter to R. F. Harrod, 4 July 1938". In: *The Collected Writings of John Maynard Keynes, Vol. XIV: The General Theory and After, Part II: Defence and Development*. Ed. by D. Mog-

gridge. Page reference to be verified against the printed volume. London: Macmillan for the Royal Economic Society.

Klump, R., P. McAdam, and A. Willman (2012). "The normalized CES production function: Theory and Empirics". In: *Journal of Economic Surveys* 26.5, pp. 769–799.

Labini, P. S. (1995). "Why the interpretation of the Cobb-Douglas production function must be radically changed". In: *Structural Change and Economic Dynamics* 6.4, pp. 485–504.

Lawson, T. (1985). "Uncertainty and Economic Analysis". In: *The Economic Journal* 95.380, p. 909.

Mazzucato, M. (2013). *The Entrepreneurial State: Debunking Public vs. Private Sector Myths*. London: Anthem Press.

– (2018). *The Value of Everything: Making and Taking in the Global Economy*. London: Allen Lane.

Newell, P. and D. Mulvaney (2013). "The Political Economy of the 'Just Transition'". In: *The Geographical Journal* 179.2, pp. 132–140.

Pollin, R. (2015). *Greening the Global Economy*. Cambridge, MA: MIT Press.

A Full proofs

This appendix reproduces, in full, the ex ante equivalence results of the revision (source note: `equivalence.tex`, v4). It states the common kernel, derives the demand-only reduction, proves the exact equivalences of the closure rows and of the sectoral family's rigid corner, isolates what does differ across the rows, and derives the scope conditions under which the equivalences break.

A.1 Purpose and summary

Reviewer 2's critique turns on the exact conditions under which the closure rows of the evaluation matrix are or are not equivalent (`docs/DOCS_ASSESSMENT.md`, section 4.2). The executed `matrix_5x3_v10` generation (ADR-0019 together with the ADR-0020 option C endpoint; 33/33 cells, run ids `matrix_5x3-v10-*`) measures two coincidences:

- **ALPHA ≡ scalar BETA (a benchmark, not the headline BETA).** The *single-wage* elastic-supply cell ($\eta_s = 0.5$; run ids `matrix_5x3-v10-BETA-F*`) reports every metric of the full-employment row identically: real GDP, the welfare index, employment, the external account — to all eight printed decimals, with the identity wedge at $\sim 10^{-16}$. A direct comparison of stored solutions measured $\max|\Delta y| = 6.2 \times 10^{-17}$ (session of 2026-09-17). The v6 generation reproduces the v5 ALPHA/BETA cells to 6.7×10^{-16}, so the coincidence survives the endpoint

change.

- **Headline (sectoral) BETA $\not\equiv$ ALPHA.** The headline BETA row is the sectoral form of Section A.5 at $\eta_s = 0.5$ (run ids `matrix_5x3-v10-BETA-F*-etas05`). With N sectoral wages the price block is not homogeneous, prices move ($\max|p-1| = 0.1057$ in F2/F3), and the row is a distinct economy; only its rigid corner $\eta_{s,i} = 0$ coincides with another row (BF).
- **GAMMA $\equiv$ DELTA.** The Leontief-corner row ($(\theta_\delta, \epsilon_\delta, \sigma_\delta) = (10^{-4}, 10^{-4}, 10^{-4})$) reports every metric of the fixed-wage row at the baseline CES elasticities $(\theta, \epsilon, \sigma) = (0.5, 0.5, 0.9)$ identically, in all three financing columns; the flow-table wedges agree at the residual level ($\sim 10^{-12}$). The analytic `leontief_multiplier` test reproduces the nonlinear DELTA solution with relative error 0.0 (recorded in `docs/DOCS_ASSESSMENT.md` section 4.2).

Sections A.2–A.3 set up the common kernel and derive the demand-only reduction. Section A.4 states and proves the two equivalences and classifies them. Section A.5 records a third exact equivalence, of a different kind: the rigid corner of the sectoral-labour-market family is the $\eta = 0$ endpoint itself, and the family interpolates between the matrix's two rigidity rows. Section A.6 isolates what *does* differ across the rows (the scalar closure regimes). Section A.7 derives the scope conditions: which model changes break the equivalences — the menu behind the demand-sensitive-price program — and which cannot.

Throughout, "demand-only shock" means the matrix design: a programme bundle $g = G_0\,\psi$ ($\sum_i \psi_i = 1$; $G_0 = 40\,300$ EUR m, ADR-0024) entering final demand under one of the three financing closures, with the supply shock at $A \equiv 1$ (unit productivity).

A.2 The common kernel

Let N be the number of sectors. Prices p_i, wage w, gross output y_i, employment L_i. Data-calibrated objects: factor shares α_i (labour), domestic bill coefficients $a_u = A_u^{bill}/\lambda_u$ with input-share matrix Ω (Ω_{iu} = the share of input u in sector i's input bundle; rows sum to one), CPI weights ω_i ($\sum_i \omega_i = 1$), sector import margins m_i, saving rate s, exogenous government demand g^G, investment inv, exports X (no margin), the programme bundle g, and the labour bar $\bar{L} = \sum_i \bar{L}_i = 1$ (baseline income units, GDP = 1). Elasticities: intermediate substitution θ, labour–intermediate substitution ϵ, consumption substitution σ, allocation $\eta \in \{0, 1\}$ (ADR-0010), supply elasticity $\eta_s \geq 0$ (BETA only).

Price block. The CES intermediate price index and unit cost are

$$P_i(p) = \Big(\textstyle\sum_u \Omega_{iu}\, p_u^{1-\theta} \Big)^{\frac{1}{1-\theta}}, \tag{A1}$$

$$c_i(p, w; A) = \Big(A_i^{\epsilon-1} \big(\alpha_i\, w^{1-\epsilon} + (1-\alpha_i)\, P_i^{1-\epsilon} \big) \Big)^{\frac{1}{1-\epsilon}}, \tag{A2}$$

with the Cobb–Douglas limit $c_i = w^{\alpha_i} P_i^{1-\alpha_i} / A_i$ at $\epsilon = 1$. Zero profit requires

$$p_i = c_i(p, w; A) \qquad \forall\, i. \tag{A3}$$

At the calibration point $(p, w, A) = (\mathbf{1}, 1, \mathbf{1})$ the unit cost equals one exactly, because $\sum_u \Omega_{iu} = 1$ makes $P_i = 1$; this is the calibration identity used throughout.

Quantity block. With full cost-minimizing allocation ($\eta = 1$) the marginal-product condition inverts to

$$L_i = \big(p_i\, A_i^{(\epsilon-1)/\epsilon}\, \alpha_i^{1/\epsilon}\, y_i^{1/\epsilon} / w \big)^{\epsilon} = p_i^{\epsilon}\, A_i^{\epsilon-1}\, \alpha_i\, y_i / w^{\epsilon}, \tag{A4}$$

At $\eta = 0$ the allocation is the frozen baseline, $L_i = \bar{L}_i$; under ADR-0020 option C the wage w in (A4) is then a per-sector w_i, solved from $L_i^{cm}(p, y_i, w_i) = \bar{L}_i$ jointly with (A3). Intermediate demand is charged with the *domestic* bill (ADR-0012):

$$int_i = p_i^{-\theta} \sum_u \Omega_{ui}\, p_u^{\epsilon} A_u^{\epsilon-1} P_u^{\theta-\epsilon}\, a_u\, y_u. \tag{A5}$$

Household demand is CES with CPI deflation and the financing weights ds ($ds = 1$ except under F1, where $ds = d$, the preference tilt):

$$c_i = (1-s)(1-m_i)\, \omega_i\, ds_i\, E\, p_i^{-\sigma} \Big/ \textstyle\sum_j \omega_j ds_j\, p_j^{1-\sigma}, \qquad E = (1-\tau)\, w \textstyle\sum_i L_i + F, \tag{A6}$$

with the lump-sum tax rate $\tau = p{\cdot}g^G / (w \sum_i L_i)$ under F1/F3 and $\tau = p{\cdot}(g^G + g) / (w \sum_i L_i)$ under F2, and F the net external transfer entering expenditure after tax (ADR-0019). At $\eta = 0$ the wage bill is $\sum_i w_i \bar{L}_i$ in (A6). Saving sE leaks; the import content is charged margin by margin. Goods-market clearing reads

$$y_i = int_i + c_i + (1-m_i)\big(g_i^G + inv_i + g_i \big) + X_i, \tag{A7}$$

where g_i appears only under the additive financings F2/F3 ($g_i = 0$ under F1; legacy manna is zero in the matrix).

The four regimes. The mobile systems solve for $X = (p, y, w, F) \in \mathbb{R}^{2N+2}$ with the numeraire $\Pi(p) = 1$, where

$$\Pi(p) = \left(\sum_i \omega_i \, p_i^{1-\sigma} \right)^{\frac{1}{1-\sigma}}, \tag{A8}$$

and close with the labour equation

$$\text{ALPHA:} \quad \sum_i L_i(p, y, w) = \bar{L}, \tag{A9}$$

$$\text{BETA:} \quad \sum_i L_i(p, y, w) = \bar{L} \left[\frac{w/\Pi(p)}{w_0/\Pi_0} \right]^{\eta_s}, \qquad w_0/\Pi_0 = 1. \tag{A10}$$

The fixed-wage systems solve for $X = (p, y) \in \mathbb{R}^{2N}$ with the wage pinned $w \equiv 1$ and $F \equiv 0$; employment is a post-solve outcome. GAMMA runs them at the baseline elasticities, DELTA at the Leontief corner $(\theta_\delta, \epsilon_\delta, \sigma_\delta) = (10^{-4}, 10^{-4}, 10^{-4})$. BF is the sectoral-wage system at $\eta = 0$ (ADR-0020 option C, superseding the ADR-0019 $F = 0$ pin): the allocation stays frozen at the baseline, $L_i = \bar{L}_i$, and N sectoral wages w_i are solved from each sector's own first-order condition $L_i^{cm}(p, y_i, w_i) = \bar{L}_i$ jointly with zero profit, with unknowns $[\, p;\ y;\ w(1{:}N);\ F \,]$ and the external transfer F free. All rows in the matrix use $\eta = 1$ except BF.

A.3 The demand-only reduction

Lemma 1 (Price block demand-invariance)**.** *The zero-profit system* (A3) *is homogeneous of degree one in (p, w) jointly and contains no demand variable. Consequently, for any technology A, every solution satisfies $p = w\,\pi(A)$, where $\pi(A) \in \mathbb{R}^N_{++}$ solves $c_i(\pi, 1; A) = \pi_i$ for all i; relative prices and the ratio p/w are functions of A alone. Under the CPI numeraire, $w = 1/\Pi(\pi(A))$ and $p = \pi(A)/\Pi(\pi(A))$; under the fixed wage, $w = 1$ and $p = \pi(A)$. At $A \equiv \mathbf{1}$ the calibration identity gives $\pi = \mathbf{1}$, hence in both gauges*

$$p = \mathbf{1}, \qquad w = 1, \qquad \Pi = 1, \tag{A11}$$

independently of the demand vector, of the financing closure, and of the shock size. The lemma is a statement about the single-wage *price block: with a per-sector wage vector (the $\eta = 0$ endpoint, ADR-0020) homogeneity in (p, w) no longer holds sector by sector and the argument does not apply. The statement is scoped to the branch the numeraire selects: $\pi(A)$ denotes the normalised solution the kernel's solves return, and the equivalences below compare the systems on that branch (the CES*

price system at $\theta < 1$ with self-loops admits spiral branches in principle; no executed cell visits one).

Proof. Homogeneity: $c_i(tp, tw; A) = t\, c_i(p, w; A)$ for $t > 0$, by inspection of (A2)–(A1). Write $q = p/w$. Then (A3) is equivalent to $q_i = c_i(q, 1; A)$ for all i, i.e. $q = \pi(A)$, and the solution branch of (A3) selected by the numeraire is the ray $\{(w\,\pi(A), w)\}$. The CPI (A8) is homogeneous of degree one in p, so $\Pi(w\,\pi) = w\,\Pi(\pi)$; the numeraire $\Pi = 1$ or the pin $w = 1$ selects the scale. At $A \equiv \mathbf{1}$, $\pi = \mathbf{1}$ solves (A3) at $w = 1$ because $c_i(\mathbf{1}, 1; \mathbf{1}) = 1$ (the calibration identity), and $\Pi(\mathbf{1}) = 1$ because $\sum_i \omega_i = 1$. □

Corollary 1 (Real-wage invariance)**.** *At every solution of the shared block,*

$$\frac{w}{\Pi(p)} = \frac{1}{\Pi(\pi(A))}, \tag{A12}$$

a function of technology alone. At $A \equiv \mathbf{1}$ this equals $1 = w_0/\Pi_0$: the equilibrium real wage sits at its anchor for every demand vector, every financing closure and every shock magnitude. (Measured in the matrix: the GDP deflator reads 1.000000 *in the twelve $\eta = 1$ cells, and a demand shock of ten times the programme size moves* $\max|p - \mathbf{1}|$ *by* 2.7×10^{-15}*;* `docs/ETAs.md`*.)*

Lemma 2 (Elasticity-free affine quantity block)**.** *Fix $\eta = 1$ and $A \equiv \mathbf{1}$, and evaluate the quantity block at the demand-only prices $(p, w) = (\mathbf{1}, 1)$. Then for* every *value of $(\theta, \epsilon, \sigma)$:*

$$int = \Omega^\top \operatorname{diag}(a)\, y, \qquad L = \operatorname{diag}(\alpha)\, y, \qquad \sum_i L_i = \alpha^\top y, \tag{A13}$$

$$c = (1 - s)\operatorname{diag}(1 - m)\,\tilde{\omega}(ds)\; E, \qquad E = \alpha^\top y - \textstyle\sum g^G + F \;\; (F1/F3),$$

where $\tilde{\omega}(ds) = (\omega \odot ds)\,/\,\mathbf{1}^\top(\omega \odot ds)$ and $E = \alpha^\top y - \sum g^G - \sum g$ under F2. Hence the clearing system (A7) *is exactly affine,*

$$y = G\,y + b(F), \qquad G = \Omega^\top \operatorname{diag}(a) + (1 - s)\operatorname{diag}(1 - m)\,\tilde{\omega}(ds)\,\alpha^\top, \tag{A14}$$

$$b(F) = (1 - m) \odot \big(g^G + inv + g\big) + X + (1 - s)\operatorname{diag}(1 - m)\,\tilde{\omega}(ds)\big(F - \textstyle\sum g^G - \mathbf{1}_{F2} \sum g\big), \tag{A15}$$

with g present only under the additive financings F2/F3 and $\mathbf{1}_{F2}$ selecting the tax-financed column (the bracket is $F - \sum g^G$ under F1/F3 and $F - \sum g^G - \sum g$ under F2). No elasticity parameter enters G or b. At $\eta = 0$ the aggregate is still pinned, $\sum_i L_i = \bar{L}$, but the unit-price reduction below describes only the retired *common-wage pin: there E did not depend on y and the household round*

dropped out, $G_{BF} = \Omega^\top \mathrm{diag}(a)$*. Under the ADR-0020 sectoral-wage closure the wages* w_i *move with the demand composition, the prices leave the unit point, and this affine form does not describe the* $\eta = 0$ *row.*

Proof. At $p = w = A = \mathbf{1}$: $P_u = 1$ in (A5), so all elasticity factors p^ϵ, $P^{\theta-\epsilon}$, $p^{-\theta}$ equal 1 and (A5) reduces to the fixed-coefficient round $\sum_u \Omega_{ui} a_u y_u$; (A4) reduces to $L_i = \alpha_i y_i$ since every exponent collapses at unit prices; in (A6) the terms $p_i^{-\sigma}$ and the normalizer reduce to 1 and $\sum_j \omega_j ds_j$, giving the fixed spending weights $\tilde{\omega}(ds)$; and the tax rate $\tau = \sum g^G / \alpha^\top y$ turns E into the affine form shown. Collecting terms in (A7) gives (A14)–(A15): the intermediate round contributes the columns $\sum_i \Omega_{ui} a_u = a_u$ (row sums of Ω equal 1) and the household round contributes $(1-s)\sum_i (1 - m_i)\tilde{\omega}_i(ds)\, \alpha_u$ per unit of y_u. □

Remark 1 (Nonsubstitution)**.** *Lemma 1 and Lemma 2 are the production side of the Samuelson nonsubstitution theorem: with one primary factor, constant returns and no joint production, the cost-minimizing technique — and hence relative prices — is independent of the demand vector, and final demand allocates quantities only (Samuelson 1951). The kernel's nested-CES structure is a smooth parametrisation of that corner; the demand-only matrix therefore inherits the theorem's rigidity, and this is what* `docs/DOCS_ASSESSMENT.md` *section 4.2 records as the "model-class property" behind the coincidences. The IO endpoint makes the rigidity literal — the Robinson (2006) fixed-price multiplier — but, as Proposition 2 shows, it is not an approximation: the same candidate equilibrium* $(p, y) = (\mathbf{1}, y^*)$ *solves the fixed-wage system at every elasticity triple in the fixed-wage gauge.*

A.4 The two equivalences

Proposition 1 (ALPHA ≡ scalar BETA)**.** *Let* $A \equiv \mathbf{1}$ *and* $\eta = 1$*. The ALPHA system* (A3),(A7),(A9),$\Pi = 1$ *and the BETA* single-wage *(scalar) BETA system* (A3),(A7),(A10),$\Pi = 1$ *have identical solution sets, for every* $\eta_s \geq 0$ *and every financing closure.*

Proof. The two systems differ in exactly one equation. By Lemma 1 and Corollary 1, every solution of the shared block satisfies $w/\Pi = 1 = w_0/\Pi_0$, so BETA's labour equation (A10) reduces to $\sum_i L_i = \bar{L}$, which is (A9). Conversely, any solution of the ALPHA system satisfies BETA's equation by the same substitution. The systems are therefore equivalent, equation by equation. □

Proposition 2 (GAMMA ≡ DELTA)**.** *Let $A \equiv \mathbf{1}$, $\eta = 1$, $F \equiv 0$, $w \equiv 1$. Then $(p, y) = (\mathbf{1}, y^*)$ with $y^* = (I - G)^{-1} b(0)$ solves the fixed-wage system at* every *elasticity triple $(\theta, \epsilon, \sigma)$; the employment outcome is $L^* = \alpha^\top y^*$, the consumption block is $c^* = (1-s)\,\mathrm{diag}(1-m)\,\tilde{\omega}(ds)\,(L^* - \sum g^G - \sum g\,\mathbf{1}_{F2})$, where $\mathbf{1}_{F2}$ selects the tax-financed column, and the block decomposition* (A13) *holds verbatim. In particular GAMMA (baseline elasticities) and DELTA (the Leontief corner) share this solution, and the analytic system* (A14)–(A15) — *`leontief_multiplier` in `src/closures/labor/labor.jl` — computes it exactly.*

Proof. Existence and form: at $p = \mathbf{1}$ the price block holds by the calibration identity (Lemma 1), and by Lemma 2 the clearing block at $p = \mathbf{1}$ is the affine system, of which $y^* = (I - G)^{-1} b(0)$ is the solution. Uniqueness on the admitted branch: the scale-determinacy guard (ADR-0017) verifies at the reference prices that $(I - G)$ is nonsingular — contraction if $\max_u \mathrm{colsum}_u(G) < 1$, otherwise a direct rank test — and that the candidate y^* is strictly positive; the guard admits the branch on which the affine fixed point is unique and positive, but does not by itself exclude further roots of the nonlinear system; the multi-init admissibility checks recorded for the F1 column (three distinct starting points converging to the same root) are the evidence that the reported root is the admitted one. Both the GAMMA and the DELTA solve are subject to the same guard, and both converged to this root (residuals 4.4×10^{-16} to 2.0×10^{-13} across the six fixed-wage cells, `runs/matrix_5x3-v10-{GAMMA,DELTA}-*/manifest.toml`). The elasticity-independence is exact in the equations, not an ϵ-limit artefact: the DELTA solve at $\epsilon_\delta = 10^{-4}$ reproduces the analytic y^* with relative error 0.0 (F2/F3 verification, recorded in `docs/DOCS_ASSESSMENT.md` section 4.2), and the v10 flow metrics of the two rows agree to all printed decimals with residual-level wedge differences $\sim 10^{-12}$. □

Corollary 2 (Classification)**.** *For the manuscript's framing of the matrix:*

Coincidence	*Status*	*What is unidentified*
GAMMA $\equiv$ *DELTA*	*definitional in form, substantive in mechanism (Lemma 2)*	$(\theta, \epsilon, \sigma)$ *in the fixed-wage gauge*
ALPHA $\equiv$ *scalar BETA*	*derivable (Cor. 1)*	η_s *in the single-wage gauge*
F2 $\equiv$ *F3 (mobile rows)*	*derivable (Section A.6)*	*the financing label;* $F_{F3} = F_{F2} - B_{gov}$
BF vs ALPHA	*not a coincidence: ADR-0020 option C gives the frozen allocation a price channel, so it is a distinct economy*	*— (the* $\eta = 0$ *row is an identified equilibrium: own prices, wage vector* 0.97–1.67*, own position)*
sectoral corner $\equiv$ *BF* ($\eta_{s,i} = 0$*; Prop. 3)*	*exact, by construction (Prop. 3)*	*— (the corner* is *the endpoint: the same equations)*

The first three entries are not numerical accidents; they hold at machine precision in the executed generation, and none is a mis-specification. They are properties of the experiment*: a demand-only programme in a one-factor CRS economy cannot identify any elasticity of the kernel (the mechanism is Remark 1). The fourth entry is different in kind: the BF/ALPHA coincidence held only in the superseded* $F = 0$ *pin generation (ADR-0019 D2), where the frozen allocation had no channel to prices; under ADR-0020 option C the* $\eta = 0$ *row carries sectoral wages and is an identified equilibrium, so no BF quantity is left unidentified and none must be withheld. The fifth entry is an equivalence between two* closures *rather than between two rows of the executed matrix, and it is exact by construction: the sectoral family's rigid corner reproduces the* $\eta = 0$ *endpoint, which is why no separate rigid sectoral cell is reported.*

Remark 2 (Off the demand-only slice)**.** *The equivalences fail the moment technology moves. At* $A \neq \mathbf{1}$*, Corollary 1 gives* $w/\Pi = 1/\Pi(\pi(A)) \neq 1$ *in general, and BETA's labour equation reduces to* $\alpha^\top y = \bar{L}\,[1/\Pi(\pi(A))]^{\eta_s}$*:* *employment is pinned to a* technology-determined constant *that now depends on* η_s*. The supply-side pilot (+20% in sector 1) measures exactly this separation — the failure of Proposition 1's hypothesis* $A \equiv \mathbf{1}$*:* $w = 1.004903$*, with* $L = 1.0024$ *at* $\eta_s = 0.5$ *versus* $L = 1.0098$ *at* $\eta_s = 2$ *against ALPHA's* $L = 1.0000$ *(`docs/ETAs.md`). The identification of* η_s

therefore requires a supply-side scenario — the premise of the supply arm.

A.5 The sectoral family and its rigid corner

The two equivalences above are coincidences *within* the executed matrix. This section records a third, of a different kind: an equivalence between two *closures*, one of which is the matrix's $\eta = 0$ endpoint and the other a corner of a family that extends beyond the matrix. It is exact rather than asymptotic, and it is what makes the labour door of Section A.7 concrete.

The family. Replace the single aggregate supply equation (A10) by N sectoral labour markets,

$$L_i^{cm}(p, y_i, w_i) \;=\; \bar{L}_i \left[\frac{w_i}{\Pi(p)}\right]^{\eta_{s,i}}, \qquad i = 1, \ldots, N, \qquad \eta_{s,i} \geq 0, \tag{A16}$$

solved jointly with zero profit (A3), clearing (A7), the numeraire $\Pi(p) = 1$ and a free transfer F, in the unknowns $X = (p, y, w(1{:}N), F) \in \mathbb{R}^{3N+1}$ (ADR-0022). The scalar closure (A10) is the member of this family with one common wage and $\eta_{s,i} \equiv \eta_s$; it is retained only as a single-wage benchmark. The matrix's BETA row is the sectoral member of the family at $\eta_{s,i} \equiv 0.5$. Because the N wages enter the N zero-profit conditions (A3) through the price index (A1), the price block now depends on the N employment levels and Lemma 1 no longer applies: demand reaches prices through the wage structure, $p = \pi(A, w)$.

Two rigidities, one equation. In both closures labour is immobile, but for different reasons. At the $\eta = 0$ endpoint the allocation is frozen *by assumption*: $L_i = \bar{L}_i$ is a datum of the quantity block, and the sectoral wage is whatever clears cost-minimizing demand against it. At $\eta_{s,i} = 0$ each sector has a *vertical* supply curve, $L_i^s = \bar{L}_i$ for every w_i, so the market clears at the same allocation with a demand-determined wage. A constraint imposed on a quantity and an infinitely inelastic price response turn out to be the same row of algebra — which is the content of the proposition below.

Proposition 3 (The rigid corner is the $\eta = 0$ endpoint)**.** *Set $\eta_{s,i} = 0$ for every i. Then the sectoral system* (A16) *and the $\eta = 0$ endpoint of ADR-0020 option C have the same solution set.*

Proof. Write both systems in the unknowns $X = (p, y, w, F)$ as four row blocks:

$$\begin{aligned}
&\text{(P)} \quad p_i = c_i(p, w_i; A), && i = 1, \dots, N, && \text{(A3)}\\
&\text{(Y)} \quad y_i = int_i + c_i + (1 - m_i)(g_i^G + inv_i + g_i) + X_i, && i = 1, \dots, N, && \text{(A7)}\\
&\text{(W)} \quad \log L_i^{cm}(p, y_i, w_i) - \log \bar{L}_i - \eta_{s,i} \log \frac{w_i}{\Pi(p)} = 0, && i = 1, \dots, N,\\
&\text{(E)} \quad \Pi(p) = 1.
\end{aligned}$$

The quantity block (Y) draws labour income from the wage bill $\sum_i w_i L_i$ inside (A6) and from the tax rate τ; this is the only place, besides (W), where the two systems differ. The $\eta = 0$ endpoint is the same four blocks with two changes: (W) drops the supply term ($\eta_{s,i} = 0$), and (Y) evaluates the wage bill at the *frozen* allocation, $L_i := \bar{L}_i$, instead of at the cost-minimizing demand L_i^{cm}.

(i) Let X solve the sectoral system at $\eta_{s,i} \equiv 0$. Its (W) gives $L_i^{cm}(X) = \bar{L}_i$ for every i, the supply term vanishing identically. Substituting $L_i^{cm} = \bar{L}_i$ into its (Y) makes that block equal the endpoint's (Y); (P) and (E) are identical in the two systems by inspection. Hence X solves the endpoint system.

(ii) Let X solve the endpoint system. Its (W) gives $L_i^{cm}(X) = \bar{L}_i$, which satisfies the sectoral (W) at $\eta_{s,i} \equiv 0$; substituting it into (Y) yields the sectoral (Y). Hence X solves the sectoral system. Both systems are square — $3N + 1$ unknowns, $3N + 1$ equations — and their residual maps agree on the common solution set, so the solution sets coincide. □

Remark 3 (Why the family interpolates, and where it does not close)**.** *Sweeping $\eta_{s,i}$ moves the adjustment from quantities to prices. At $\eta_{s,i} = 0$ the allocation is pinned and the N wages absorb the shock; as $\eta_{s,i} \to \infty$ the supply curves become infinitely elastic, the real wage returns to its technology-determined anchor and employment becomes the absorbing margin — GAMMA-like prices and GAMMA's absorbing margin. The two ends of the family are therefore the matrix's own rigidity rows: BF (allocation frozen, wages free) at the rigid corner, GAMMA (real wage pinned, allocation free) at the elastic one. Two qualifications are exact and matter for reading the table below. First, the elastic end is a* limit*, not a cell: the supply slope diverges and the direct solve eventually fails ($\eta_s = 10^6$; probe13). Second, the family keeps the transfer F free, whereas the fixed-wage system pins $F \equiv 0$; the family therefore* spans *the two rigidity rows without terminating in either.*

Measured. Table A1 reports the executed sectoral cells (`matrix_5x3-v10-BETA-*`; ADR-0022). Four readings.

(a) Prices move, and with the demand composition. Every sectoral cell carries a deflator different from one and a $\max|p-1|$ that differs between F1 and F2/F3 — the signature no single-wage closure produces (all twelve $\eta = 1$ cells of the matrix read exactly 0.000000). *(b) The elasticity splits the adjustment.* As η_s rises from 0.25 to 2, the price response falls monotonically (0.1529 $\rightarrow$ 0.0372 at F2) while employment rises (1.000864 $\rightarrow$ 1.001914): the family trades price adjustment for quantity adjustment, and the welfare cost falls with it ($-1.69\% \rightarrow -1.39\%$). The F1 column, whose programme is financed by a preference tilt rather than a tax, is the exception: its welfare index is positive (+0.42% at $\eta_s = 0.5$). *(c) Incidence matters more than the level.* Making the *programme* sectors rigid ($\eta_{s,i} = 0$ there, $\eta_s = 0.5$ elsewhere) more than doubles the price response (0.2720 against 0.1057) and *reverses* the employment effect: employment falls by 0.34% and the welfare index by 2.69%, against +0.13% and −1.57% for the uniform cell. Rigidity on the largest half by baseline employment instead gives 0.1234 with a near-neutral +0.02%. *(d) Wage dispersion is large.* The mechanism moves relative wages: $\max_i w_i / \min_i w_i$ reads 1.26 at $\eta_s = 0.5$ and 1.70 for the rigid-programme variant, against 1.72 at the $\eta = 0$ endpoint — for a programme worth 1.33% of GDP.

Table A1: The sectoral family in the `matrix_5x3_v10` generation (`matrix_5x3-v10-BETA-*`; deviations from the no-programme baseline; F2 and F3 agree to all printed decimals in every cell).

Sectoral cell	$\max\|p-1\|$ (F1)	$\max\|p-1\|$ (F2)	Employment (F2)	Consumption (F2)
uniform $\eta_s = 0.25$	0.133610	0.152897	1.000864	−0.016888
uniform $\eta_s = 0.5$	0.095839	0.105666	1.001255	−0.015712
uniform $\eta_s = 1$	0.061283	0.065422	1.001627	−0.014669
uniform $\eta_s = 2$	0.035629	0.037171	1.001914	−0.013912
rigid programme, $\eta_s = 0.5$	0.215061	0.272033	0.996621	−0.026914
rigid largest half, $\eta_s = 0.5$	0.109583	0.123362	1.000196	−0.019053
BF ($\eta_{s,i} = 0$)	0.221442	0.279009	1.000000	−0.019800
GAMMA (fixed wage)	0.000000	0.000000	1.001260	−0.015333

The corner is the endpoint, numerically. Solving the sectoral system at $\eta_{s,i} = 0$ from the $\eta = 0$ solution reproduces (p, y, w, F) exactly — $\max|\Delta| = 0.00 \times 10^0$ in all four blocks, and below 10^{-10} from a cold start (`tests/test_sectoral_labour.jl`; ADR-0022) — as Proposition 3 requires. No separate rigid sectoral cell is therefore reported: the corner *is* the BF row of Table A2.

The proof never uses $A = 1$. Nothing in the four row blocks of the proposition's proof refers to the technology shock, so the nesting is not a demand-only property. Verified directly: under a +20% productivity shock in sector 1 the canary again reads 0.00×10^0 in all four blocks

(`probe15_sectoral_supply.jl`; the figures are recorded in `docs/log/2026-09.md` and `docs/ETAs.md`). The same probe shows why $\eta_{s,i}$ matters outside the demand-only slice: the rungs separate under the shock (employment 1.000000, 1.002187, 1.008218 and the real-wage spread 1.215, 1.105, 1.043 at $\eta_s = 0, 0.5, 2$), so the sectoral elasticity is identifiable only with a supply-side scenario, while under a demand-only programme it is a sensitivity parameter.

A.6 What actually differs across the rows

Lemma 2 shows that the demand-only matrix is a partition of *scalar closure regimes* over one and the same linear system. Two scalar regimes occur across the $\eta = 1$ rows, and the $\eta = 0$ endpoint is now a third, genuinely nonlinear regime (the sectoral-wage system):

Rows	F	Employment	Household round in G
ALPHA, BETA ($\eta = 1$)	free (instrument)	pinned, $L = \bar{L}$	active
GAMMA, DELTA ($\eta = 1$)	$F \equiv 0$	endogenous, $L = \alpha^\top y$	active
BF ($\eta = 0$)	free (solved)	pinned via allocation	active (wage bill)

The F-instrument. In the mobile regime the labour pin selects the transfer. Writing $y(F) = (I - G)^{-1}(b_0 + f\,F)$ with $f = (1 - s)\,\mathrm{diag}(1 - m)\,\tilde{\omega}(ds)$, the pin $\alpha^\top y = \bar{L}$ gives

$$F^* = \frac{\bar{L} - \alpha^\top (I - G)^{-1} b_0}{\alpha^\top (I - G)^{-1} f}. \tag{A17}$$

This instrument is why the mobile F2 and F3 columns are real-neutral: F2 charges the programme through the lump-sum tax ($E = \alpha^\top y - \sum g^G - \sum g$), F3 through the external account ($E = \alpha^\top y - \sum g^G + F$), and the two b-vectors differ by exactly $f \sum g$. The pin (A17) therefore yields $F^*_{F3} = F^*_{F2} - \sum g$, identical real allocations, and an identical net external position $F + B_{gov}$. Measured: $F_{F3} - F_{F2} = -0.01330991 = -G_0$ to eight decimals in every mobile row (the F values are printed in `paper/superseded/matrix_5x3_v6_flows.md`, Table 2, which the v10 re-mint reproduces to all printed decimals; $G_0 = \sum_i g_i = 0.013310$, ADR-0024).

Who pays. The economically meaningful difference between the two paradigms is which margin absorbs the programme. In the mobile rows the labour pin forces the household to pay (consumption -1.82% under F2/F3; the F instrument crowds it out exactly), while real GDP is flat by construction ($w\,\bar{L}$ deflated, ADR-0018). In the fixed-wage rows employment is the margin: F3 expands employment by 1.78% and consumption by 2.26% (the Keynesian case), F2 leaves GDP nearly

neutral (+0.13%) with a welfare loss from the tax, and F1 shifts composition slightly downward (−0.10% employment). The $\eta = 0$ row is no longer degenerate: at the sectoral-wage endpoint the welfare index reads +0.096% (F1) and −1.980% (F2/F3), the deflator moves (1.0063 / 1.0072), and the external account closes in every column. Two caveats from the ADR-0020 audit now describe the *superseded* v5 pin generation: there the recorded BF/ALPHA differences were carried by the pin, not by the frozen allocation (replacing the pin by the labour equation reproduced the ALPHA cells to 2.8×10^{-17}), and the $\eta = 0$ external position was not identified (every pin value $F = c$ an exact root that moved the booked position one-for-one). Option C resolves both: the account closes to $\leq 1.04 \times 10^{-11}$ at BF, F is a solved equilibrium object, financing neutrality extends to $\eta = 0$ ($F_{F3} = F_{F2} - B_{gov}$ to 3.5×10^{-14}), and no BF entry need be withheld.

Admissibility. All fifteen cells execute (v6 and, re-minted, v10). In particular the F1 column is admissible in the fixed-wage rows: the saving rate and the import margins supply the marginal leakage that keeps $\max \mathrm{colsum}(G) < 1$ (ADR-0017's actual-matrix criterion reads 0.8314 on full-71 and 0.8320 under the F1 tilt); the endogenous tax rate $\tau = \sum g^G / \alpha^\top y$ is a fixed-revenue level ($T = p \cdot g^G$) and enters b, not G (the retired heuristic guard had blocked these cells in the v1 generation).

A.7 Scope conditions: what breaks the equivalences

Both propositions rest on Lemma 1: the price block contains no demand variable. Any model change that lets demand into that block restores identification and creates variety in the published matrix. The doors are:

1. **Labour door — sector-specific wages.** With N sectoral labour markets $w_i = \phi_i(L_i)$ instead of one common w, the homogeneity argument of Lemma 1 collapses: $L_i = \alpha_i y_i$ depends on demand, so w_i and through zero profit p_i become demand-sensitive. Zero profit is preserved (no rents), the new parameters are sectoral supply elasticities of the same family as η_s, and the variant repairs the BF row's common-wage defect (the `labor.BF` open gate of `registry/closures.toml`, resolved by ADR-0020 option C; `cbase2/review.md` section 1). Ranked candidate 1 in `docs/ETAs.md`. ADR-0020 option C is this door, adopted at the $\eta = 0$ endpoint: it is why the v6 BF row is a distinct economy rather than a normalisation of ALPHA. ADR-0021 (proposed) reuses the vector-wage apparatus in the fixed-wage row, but with an *exogenous* pin — that delivers design heterogeneity, not demand-sensitivity, because the

pinned vector keeps the price block demand-free. ADR-0022 (accepted) generalises this door across the $\eta = 1$ rows: N sectoral labour markets with supply elasticities $\eta_{s,i}$ (Section A.5), whose rigid corner is precisely the option C endpoint proved in Proposition 3. It breaks the single-wage homogeneity on which Lemma 1 rests, requires no income closure (no rents are created), and its scalar member is the existing BETA row.

2. **Markup door — demand-sensitive markups.** A pricing rule $p_i = \mu_i(y_i)\, c_i(p, w)$, e.g. $\mu_i = 1 + \kappa\,(y_i/y_{i0} - 1)$, breaks Lemma 1 directly. The real cost is not code but accounting: markup income is profit, so a profit-income-and-spending closure is required or the external-account identity behind the canary breaks. Nested at $\kappa = 0$. Candidate 2 in `docs/ETAs.md`.

3. **Capacity door — a fixed factor, or its reduced form.** A sector-fixed factor with a rental schedule makes marginal cost rise in y_i (rents require a rental-income closure — the largest surgery, candidate 3 in `docs/ETAs.md`). The reduced-form variant avoids all income-closure problems: an endogenous utilization externality $A_i^{eff} = A_i\,(y_i/\lambda_i)^{-\delta}$ raises unit cost with demand while keeping zero profit (the extra cost is paid to no one), so a one-line change in the cost hook (A2) suffices. With the opposite sign ($\delta < 0$) it is the Kaldor–Verdoorn case: increasing returns make prices *fall* with demand — heterodox-friendly variety in the other direction. (New candidate; see Section A.8.)

4. **External door — endogenous import prices.** An upward-sloping foreign supply curve $p_i^M = p_{i0}^M\,(M_i/M_{i0})^{\zeta}$, $\zeta \geq 0$, enters the intermediate index (A1) and the CPI through the margins, so π becomes a function of import volumes and hence of demand. The surgery is moderate and it exploits precisely the external-account machinery the paper has just built (ADR-0019): the programme's import content would bid up import prices — a structuralist absorption/term-of-trade channel that fits the open-economy framing. (New candidate.)

5. **Technology door — supply shocks.** $A \neq 1$ breaks Corollary 1 and separates BETA from ALPHA (Remark 2); this is the supply arm already designed in `docs/ETAs.md`, not a repair of the demand-only matrix.

Corollary 3 (No-go for nominal wage rules)**.** *Within the kernel's one-factor CRS price block, no rule that writes the* nominal *wage as a function of endogenous aggregates — an aggregate wage curve* $w = \bar{w}\,(\alpha^{\top} y/\bar{L})^{\varphi}$*, nominal indexation, or a re-anchored numeraire — can create demand-sensitive* real *outcomes. By Lemma 1,* $p = w\,\pi(A)$ *and* $\Pi(p) = w\,\Pi(\pi)$ *at every solution, so the real wage* $w/\Pi = 1/\Pi(\pi(A))$ *is invariant to whatever sets* w*; the rule merely re-labels the nominal scale. With*

the CPI numeraire or the $w = 1$ pin such a rule is either vacuous or over-determines the system (in the fixed gauge it forces $\alpha^\top y = \bar{L}$, i.e. silently reverts GAMMA to ALPHA). A rule on the real *wage, $w/\Pi = \phi(L)$, is BETA's equation* (A10) *rearranged and is already in the model — where it is pinned by technology under demand-only shocks. The numeraire lever is likewise dead (DE-0011). A single common wage curve is also what a* single-wage *price block can express; the sectoral-wage door above escapes the corollary precisely by breaking the single-wage homogeneity on which it rests.*

Quantity-side levers (endogenous saving rates, endogenous export demand) change G or b but not the price block; they add variety between financing columns, not between closures, and leave Lemma 1 intact.

A.8 Numerical evidence

The fifteen matrix cells were re-executed as the `matrix_5x3_v9` generation on the extended kernel that carries the sectoral family (ADR-0022), and are cited here from the re-minted `matrix_5x3_v10` generation (the C1-corrected measurement, 2026-09-20). v6 and v9 are bit-identical to each other ($\max|\Delta| = 0$ on `gdp_rel`, `consumption_rel`, employment, the deflator, $\max|p-1|$, the external transfer, the wage extremes and the identity residual); v10 reproduces them to solver precision (max 9.34×10^{-9} at the stiff BF-F1 `wage_max` diagnostic, 1.63×10^{-11} in the worst ALPHA/BETA/GAMMA/DELTA row, 4.55×10^{-15} at the exact-baseline cells; no printed figure is affected, `revision/tables/matrix_5x3_v10_flows.md`, Table 2), so the equivalences and the table below are invariant to the extension. The sectoral cells of the same generation are reported separately in Table A1.

Table A2 extracts the rows of the executed `matrix_5x3_v10` generation (the fifteen matrix cells reproduce their v6/v9 manifests to all printed decimals) (income-side real GDP and the household-consumption welfare index, ADR-0018; employment). The BETA rows are the sectoral cells at $\eta_s = 0.5$ (`matrix_5x3-v10-BETA-F*-etas05`); they are a distinct economy. The single-wage BETA cells, which coincide with ALPHA to all printed decimals, are kept as a benchmark (Appendix C). The paired rows (GAMMA/DELTA) are identical to all printed decimals in every reported metric, including the full flow decomposition (saving, tax, investment plus exports, imports, taxes on intermediates; see `paper/superseded/matrix_5x3_v6_flows.md`, whose matrix rows the v9 and v10 generations reproduce to all printed decimals). The three BF rows are included for contrast: the $\eta = 0$ sectoral-wage endpoint is a distinct economy, not a normalisation of ALPHA. The GDP deflator reads 1.000000 in the nine $\eta = 1$ ALPHA and GAMMA/DELTA cells — the numerical face of

Lemma 1 — while the three BF $\eta = 0$ cells carry a deflator of 1.006326 (F1) and 1.007181 (F2/F3), the price response the sectoral wages create.

Table A2: The matrix rows (BETA = sectoral, $\eta_s = 0.5$) in the `matrix_5x3_v10` generation (run ids `matrix_5x3-v10-*`; deviations from the no-programme baseline).

Row	Financing	Real GDP	Consumption (welfare)	Employment L
ALPHA	F1	+0.000000	+0.001432	1.000000
BETA	F1	+0.001217	+0.004227	1.001191
ALPHA	F2	+0.000000	−0.018226	1.000000
BETA	F2	+0.001287	−0.015712	1.001255
ALPHA	F3	+0.000000	−0.018226	1.000000
BETA	F3	+0.001287	−0.015712	1.001255
GAMMA	F1	−0.000973	−0.000806	0.999027
DELTA	F1	−0.000973	−0.000806	0.999027
GAMMA	F2	+0.001260	−0.015333	1.001260
DELTA	F2	+0.001260	−0.015333	1.001260
GAMMA	F3	+0.017796	+0.022644	1.017796
DELTA	F3	+0.017796	+0.022644	1.017796
BF	F1	−0.000110	+0.000963	1.000000
BF	F2	−0.000162	−0.019800	1.000000
BF	F3	−0.000162	−0.019800	1.000000

Implications for the manuscript. (i) The two-paradigm reading is safe to state as a theorem-backed claim: the flexible-wage rows form one equivalence class (composition at pinned employment), the fixed-wage rows another (employment at pinned real wage), and the difference between the classes is the F-instrument — a derivation, not an observation. (ii) The null results for η_s and for $(\theta, \epsilon, \sigma)$ under demand-only shocks should be reported as identification statements (Corollary 2), which is precisely what reviewer 2's critique needs. (iii) The variety program for the demand-only matrix must route through one of the five doors of Section A.7; within it, the capacity door's utilization/Verdoorn variant and the external door are the cheap additions beyond the three candidates already ranked in `docs/ETAs.md`, while Corollary 3 rules out the entire nominal-wage-rule family ex ante. The labour door is no longer hypothetical: ADR-0020 option C delivers it at the $\eta = 0$ endpoint, and the general sectoral-labour-market form extends it across the $\eta = 1$ rows. (iv) The labour door is now an implemented mechanism rather than a proposal: the sectoral family of Section A.5 is promoted, tested and executed (ADR-0022), it delivers demand-sensitive prices inside a flexible-wage closure, and it leaves the executed matrix unchanged to solver precision (no printed figure affected). The manuscript can therefore state three exact equivalences and present the family as the answer to the variety question — with two caveats that must travel with it: under demand-only shocks $\eta_{s,i}$ is a *sensitivity* parameter, not an estimate, so the level of the price response is identified only by the

supply arm (Remark 2); and the rule that makes a subset of sectors rigid is a modelling choice, to be reported as a band rather than a point.

B Kernel, calibration, and A-bill accounting

Reproducibility

C Matrix variants and gamma/beta representation robustness

D Vector construction and data provenance

E Closures and the neo-Keynesian mapping

This appendix maps the CGE/IO cleavage and the neo-Keynesian alternative — the EU JRC FIDELIO model — onto the closure matrix. It supports Section 3.

E.1 The core theoretical cleavage: demand shocks in CGE versus IO models

A central question in applied input–output modeling — especially for evaluating green public investment programmes such as the EU Recovery and Resilience Facility or large-scale building renovation — is whether an exogenous demand stimulus:

1. **expands real aggregate output** by mobilising unutilised resources (the Keynesian / Leontief perspective), or
2. **merely reallocates scarce resources**, causing factor prices to rise and crowding out private activity with virtually zero net change in aggregate real GDP (the canonical neoclassical CGE perspective).

Why standard CGE models produce minimal aggregate effects from demand shocks. In textbook Walrasian CGE models: (i) *full factor utilisation* — aggregate primary factor endowments $(\bar{L}, \bar{K})$ are exogenously given and fully employed; (ii) *instantaneous market clearing* — prices and wages are perfectly flexible and clear all commodity and factor markets instantaneously; (iii) *Say's*

Law operates — output is supply-determined by the production possibility frontier $Y = F(K, L, \mathbf{X})$; (iv) *crowding out* — an exogenous increase in public investment raises demand in targeted sectors, but because total labour and capital cannot expand, the extra factor demand bids up wages and capital rentals, and the relative price increase squeezes the margins of non-shocked sectors, crowding out private consumption, private investment or net exports. The aggregate real GDP response is negligible ($\Delta Y \approx 0$).

E.2 The macroeconometric alternative: the EU JRC FIDELIO model

The **FIDELIO** model (*Fully Interregional Dynamic Econometric Long-term Input–Output*; JRC / DG GROW; Rocchi et al., 2019, 2025) was built specifically to overcome this limitation. Rather than adopting a Walrasian general-equilibrium core, FIDELIO is a *dynamic macroeconometric, multi-sectoral Neo-Keynesian* model grounded in the official Eurostat/JRC FIGARO inter-country input–output tables (64 NACE industries, 46 countries/regions). It generates positive, non-trivial real output and employment multipliers from demand shocks through five structural features:

1. **Involuntary unemployment and the wage-setting curve.** Wages do not clear the labour market. The baseline economy features involuntary unemployment ($u > 0$). Nominal wages adjust sluggishly along an econometrically estimated *wage curve* (Blanchflower–Oswald tradition):

$$\Delta \ln w_{i,r,t} = f(\text{inflation}_{t-1},\ \text{productivity}_{t},\ u_{r,t-1}).$$

   Because labour supply is not vertical at baseline, firms can hire idle workers to expand production without hitting an immediate supply wall.

2. **Mark-up pricing and sluggish cost pass-through.** Product prices are set via markups over unit costs (derived from flexible Translog cost functions over capital, labour, energy and intermediate inputs) rather than jumping to Walrasian clearing levels. Output in the short-to-medium run is *demand-determined*.

3. **Keynesian multiplier loops (the ECM consumption block.)** Household consumption follows a dynamic error-correction model tied to current real disposable income rather than to an intertemporal Euler equation:

$$\Delta G \Rightarrow\uparrow Y \Rightarrow\uparrow L \Rightarrow\uparrow (w \cdot L) \Rightarrow\uparrow C^{\text{household}},$$

inducing a secondary Keynesian expenditure loop (a Type-II multiplier), yielding aggregate output multipliers between 1.2 and 1.8 in JRC policy simulations.

4. **Dynamic capital accumulation (endogenous capacity).** Capital is not a fixed static endowment; investment responds dynamically through an accelerator / user-cost formulation, so sustained demand increases capacity utilisation and expected returns, prompting firms to invest and expand capital stocks over time.

5. **Medium-to-long-run re-equilibration.** As unemployment drops and capacity tightens, the wage curve drives up wages and unit costs; the resulting rise in output prices erodes export competitiveness, generating gradual crowding out toward a medium-run NAIRU — unless the stimulus was in public capital or R&D, in which case it permanently shifts potential output outward.

E.3 Mapping into the `BeyondHulten` 5×3 closure matrix

The platform organises macroeconomic closures across two dimensions (five labour closures $\times$ three financing closures).

The five labour closures.

1. **ALPHA** (full-employment mobile CGE benchmark): $\sum_i L_i = \bar{L}$, single flexible wage w, cost-minimising intersectoral allocation ($\eta = 1$).

2. **BF** (immobile-labour endpoint selector): $\eta = 0$ freezes sectoral labour quantities to baseline shares $L_i = \bar{L}_i$; with the sectoral-wage endpoint (ADR-0020) the wage vector is free and the frozen allocation carries a price channel.

3. **BETA** (elastic labour supply on the real wage): $\sum_i L_i = \bar{L}\,[(w/P)/(w_0/P_0)]^{\eta_s}$, deflated by the CPI.

4. **GAMMA** (fixed real wage, uncapped extensive margin): $w/P = \bar{w} = 1$, employment $L = \sum_i L_i$ completely demand-determined.

5. **DELTA** (Leontief IO corner): GAMMA ($w/P = 1$) combined with the zero-substitution limit of the CES production core ($\theta, \epsilon, \sigma \to 0^+$), reproducing the fixed-price Leontief multiplier.

(Complementarity extension **ZETA**: $0 \leq \bar{L} - L \perp w/P - \bar{\omega} \geq 0$, regime-switching between GAMMA slack and ALPHA full employment; recorded as an idea, not a closure.)

The three financing closures.

1. **F1** (preference reallocation / demand tilt): budget-neutral composition shift within fixed household expenditure ($\sum_i p_i c_i = E_h$); no net fiscal injection.
2. **F2** (tax-financed public investment): balanced government budget ($\sum_i p_i g_i = T(p)$); autonomous public investment financed by an explicit domestic tax counterparty.
3. **F3** (external / debt-financed programme): autonomous public spending financed externally or via debt ($\sum_i p_i g_i = F$), with imports and external balances absorbing the deficit.

Labour row	F1 (demand tilt)	F2 (tax-financed)	F3 (debt / external)
ALPHA (full L)	sectoral reallocation	full crowding out	crowding out (w bids up)
BF (immobile)	bottlenecks / wedges	contractionary wedge	bottleneck limits
BETA (elastic)	moderate shift	mild expansion	expansion via w/P
GAMMA (fixed w)	demand-driven shift	balanced-budget multiplier	Keynesian multiplier
DELTA (IO/Leontief)	IO composition shift	Leontief balanced multiplier	pure IO multiplier

Table A3: Where the standard CGE and the FIDELIO model sit in the matrix.

Matrix location of models.

- **Where standard CGE sits:** ALPHA-F1 or ALPHA-F2 — fixed aggregate labour $\bar{L}$ and flexible wages force crowding out; real output effects are close to zero.
- **Where FIDELIO sits:** in its static / basic IO mode (modules 1–2) at DELTA-F3 (a pure Leontief input–output multiplier driven by external funding); in its short-run dynamic mode at GAMMA-F3 (fixed real wage, unconstrained extensive margin, autonomous injection), activating the full Keynesian respending multiplier; and, in the medium-to-long run, moving dynamically from GAMMA-F3 toward BETA-F3 as the wage curve drives real wage growth.

E.4 How neo-Keynesian assumptions map into our closures

In this platform, neo-Keynesian assumptions are not auxiliary parameters — they define the mathematical structure of the closure.

A. Wage rigidity (extensive margin versus wage inflation). In accordance with ADR-0014 and DE-0004, wage rigidity here is defined on the *real wage* ($w/P = \bar{w}$), deflated by the CPI numeraire ($P = 1$). In GAMMA, fixing $w/P = 1$ removes the wage as an equilibrating price: the labour-market equation changes from clearing $L = \bar{L}$ to an unconstrained demand equation

$L = \sum_i L_i(y_i, \mathbf{p})$, and the extensive margin of employment absorbs the demand shock. Asymmetric rigidity (ZETA) follows the spirit of Baqaee–Farhi (2022): downward wage rigidity creates a regime switch — demand deficits cause unemployment (GAMMA regime), while demand surges that exhaust slack trigger wage inflation (ALPHA regime).

B. Price rigidity and fixed markups. In standard New-Keynesian models, Calvo or Rotemberg pricing prevents prices from adjusting instantaneously to demand shocks. Here, under GAMMA, wages are pegged ($w = 1$) and the shock is demand-driven, so marginal costs remain roughly constant and product prices exhibit high inertia ($\mathbf{p} \approx 1$). In DELTA, the zero-substitution limit ($\theta, \epsilon, \sigma \to 0^+$) prevents firms from adjusting technical factor proportions, so cost pass-through becomes purely linear and additive, reproducing the classic fixed-price Leontief/Robinson (2006) multiplier.

C. Non-Ricardian demand and disposable-income loops. Standard CGE models enforce intertemporal optimisation in which households anticipate future tax liabilities, neutralising debt-financed spending. Here the consumption block routes household income $E_h = (1 - \tau)w \sum_i L_i +$ capital income; under F3 the absence of an immediate tax hike allows the newly generated wage income $w \cdot \Delta L$ to fuel private consumption directly, generating an endogenous Keynesian multiplier.

E.5 Summary and methodological takeaway

1. **The binary wage regime dominates.** The literature often debates the elasticity of factor substitution or intersectoral mobility (η). As shown in `definitive_guide.md` and our runs, intersectoral mobility produces second-order reallocation effects, whereas the wage regime (flexible ALPHA versus sticky GAMMA) is the first-order determinant of the aggregate multiplier.

2. **Reconciling applied discrepancies.** When policy studies using macroeconometric models (FIDELIO, E3ME) report output multipliers around 1.5 for green investments while standard CGE models report zero or negative multipliers, the divergence is not an empirical dispute over green technology. It is a direct mathematical consequence of evaluating the shock at GAMMA/DELTA-F3 (slack resources, wage stickiness, debt financing) versus ALPHA-F2 (full employment, market clearing, domestic tax financing).